A Progressive Semantic State Framework for Language Understanding

From Contextual Spectrum Tokenization **(CST)**
to Quantum Contextual Spectrum Tokenization **(QCST)**
A Unified Context-Driven Token Representation Framework

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Abstract

This paper presents a unified and progressive formulation of Contextual Spectrum Tokenization (CST) and its quantum-inspired extension, Quantum Contextual Spectrum Tokenization (QCST). The work argues that classical tokenization enforces premature semantic collapse, resulting in irreversible information loss before contextual evidence becomes available. CST reformulates tokenization as a context-conditioned semantic spectrum, while QCST further generalizes this spectrum into a quantum-inspired semantic state governed by contextual operators and measurement principles. Importantly, the proposed framework is fully executable on classical hardware, with the quantum formalism serving as a mathematical and conceptual generalization rather than a hardware requirement.

1. Introduction

Tokenization is traditionally treated as a static preprocessing step that converts raw text into discrete symbols prior to semantic reasoning. Despite major advances in neural architectures, attention mechanisms, and representation learning, tokenization has remained conceptually unchanged. This paper challenges the assumption that semantic identity must be fixed before context is observed.

We argue that meaning is not an intrinsic property of tokens, but an emergent property of interaction between
tokens and context. This insight motivates a shift from static token identifiers to dynamic semantic states.

2. Limitations of Classical Tokenization

Classical tokenization applies a deterministic mapping:

text → token IDs → embeddings → meaning

This ordering enforces early semantic commitment. Ambiguous tokens such as “bank”, “charge”, or “interest”
are assigned fixed identifiers without access to future contextual evidence. Any alternative interpretations
are implicitly discarded.

Formally, tokenization applies a many-to-one mapping $f(t) = id$, where id is context-independent. This mapping
is irreversible and violates basic principles of inference under uncertainty.

3. Contextual Spectrum Tokenization (CST)

CST introduces a fundamental change: tokens are represented as semantic spectra rather than fixed symbols.

Definition:
A token t is represented as a spectrum:
$S(t) = \{ (m_i, w_i) \}$ for $i = 1..k$

where:
- m_i denotes a potential semantic interpretation
- w_i denotes the contextual weight of that interpretation
- $\Sigma w_i = 1$

This representation preserves ambiguity explicitly and allows meaning to evolve as context accumulates.

3.1 Contextual Fusion in CST

Context in CST is modeled as a transformation operator acting on the semantic spectrum.

Let C denote a contextual signal (sentence, document, metadata, modality). Then:
$S_{t+1} = F_C(S_t)$

The fusion operator F_C redistributes semantic weights rather than selecting a single meaning. This allows progressive refinement instead of early collapse.

4. Motivation for Quantum Generalization

While CST models ambiguity probabilistically, it treats semantic components as independent. However, linguistic phenomena such as metaphor, sarcasm, and irony demonstrate non-linear interactions between meanings.

This motivates a stronger formalism where meanings are not merely weighted, but coherently combined. Quantum mechanics provides a mathematically rigorous framework for such state interactions.

5. Quantum Contextual Spectrum Tokenization (QCST)

QCST generalizes CST by representing token meaning as a quantum-inspired semantic state.

A token is represented as a state vector:
$|\psi\rangle = \Sigma\, \alpha_i\, |m_i\rangle$

where:
- $\alpha_i \in \mathbb{C}$ are complex amplitudes
- $|\alpha_i|^2$ corresponds to the probability of observing meaning m_i

Unlike classical probabilities, amplitudes can interfere constructively or destructively.

5.1 Context as a Quantum Operator

Contextual information is modeled as an operator U_C acting on the semantic state:

$|\psi'\rangle = U_C\, |\psi\rangle$

Different context sources correspond to different operators. Composition of contexts is

achieved via operator
multiplication, allowing structured and hierarchical context modeling.

5.2 Semantic Interference

QCST allows semantic interference, where combined meanings produce effects not expressible through
independent probabilities:

$$|\alpha\_i + \alpha\_j|^2 \neq |\alpha\_i|^2 + |\alpha\_j|^2$$

This provides a principled explanation for emergent semantic effects observed in natural language.

5.3 Semantic Measurement and Collapse

Final meaning selection is modeled as a measurement process:

$$m^* \sim \text{Measurement}(|\psi\rangle)$$

Collapse occurs only when required by downstream tasks, allowing maximal preservation of ambiguity until
decision time.

6. Classical Execution of QCST

Although QCST adopts quantum formalism, execution does not require quantum hardware. State vectors,
operators, and measurements are approximated using classical linear algebra on CPUs and GPUs.

This ensures deployability, reproducibility, and integration with existing NLP pipelines.

7. Conclusion

This work presents a coherent progression from CST to QCST, redefining tokenization as a semantic state
evolution problem. By preserving ambiguity, modeling context as transformation, and deferring semantic
collapse, the framework aligns more closely with human language understanding and opens

new directions for
language model design.